



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

**TECHNO COMMERCIAL
OFFER FOR COLUMN &
CONSENSERS**

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 1 OF 14

**CERTIFIED COMPANY BY THE AMERICAN SOCIETY OF MECHANICAL
ENGINEERS (ASME U STAMP)**



ISO 9001:2015 CERTIFIED COMPANY BY TUV NORD GERMANY



Plot No. E6, E-Block, Bhosari Industrial Area, Opposite NARI,
Bhosari, Pune – 411026, Maharashtra, INDIA.

**TECHNO-COMMERCIAL OFFER FOR
COLUMN & CONSENSERS**

OFFER NO.	APSPL/PSCL/1592/25-26/R0	OFFER DATE :	15-01-2026
PREPARED BY	MS. AABHA KULKARNI	CONTACT PERSON:	
CLIENT :			



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

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ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 2 OF 14

SUBJECT: TECHNO-COMMERCIAL OFFER FOR COLUMN & CONDENSERS

Dear Sir,

We thank you for your enquiry for subject equipment. Based on the data provided by you, we are pleased to submit our Technical offer for the same as below –

1) FLASH VESSEL

Tag No: FV-51208/51302/51407/51801/51805/FV51810/51102/51103/51906/51908

Process Details:	Shell	Limpet
Fluid	Client to confirm	Client to confirm
Operating Temperature (°C)	180	280
Design Temperature (°C)	250	300
Operating Pressure (bar g)	1.5/FV	6
Design Pressure (bar g)	3.5 / Full Vacuum	20/FV
Hydro Test Pressure (bar g)	4.5	26
Density (kg/ m³)	Client to confirm	Client to confirm
Corrosion Allowance	1	NIL
Radiography (Shell/Dish)	100% + 10 % + all T Joints	100% + 10 % + all T Joints
Joint Efficiency (Shell/Dish)	0.85/1	1
Vessel Details :		
Design Code	ASME SEC VIII, DIV-1	
Tank Shape	Cylindrical	
Orientation	Vertical	
Working volume (KL)	6.0 KL	
Gross Volume (KL)	9.8 KL	
Shell Inside Diameter	1800 mm	
Shell height TL to TL	3200 mm	
Shell thickness	10 mm (Nom)	
Top dish thickness	12 mm (Nom)/ 10.8 (Min)	
Bottom Dish thickness	12 mm (Nom)/ 10.8 (Min)	
Type of dish end	Tori spherical	
Limpet		



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 3 OF 14

Limpet OD mm	60	
Limpet pitch	90	
Limpet thickness mm	4	
No of turns	~ 11 on shell & ~9 Nos on dish	
Material of Construction	Shell / top dish / RF PAD	SA240GRSS316
	Body Flanges	NA
	Nozzle Pipe	SA312TP316(S)
	Nozzles Flanges up to 80NB	SA182F316
	Nozzles Flanges above 80NB	SA 516 GR 70 WITH SS316 OVERLAY
	Limpet Coil	SA240GR304
	Limpet Nozzle Pipe	SA312TP304 (S)
	Limpet Nozzle Flange	SA182F304(SORF)
	Gasket	PTFE
	Int Bolt & Nuts	SS316
	Ext Fastners	SA193GRB7/SA194GR2HWITH HDGI
	Support	IS2062GRA WITH SS304 PAD
	Lifting Lug	IS2062GRA WITH SS304 PAD
	Internal	SA240GR316
	Insulation	IN CLIENTS SCOPE
Earthing Boss	SS304 WITH BOLT	
Insulation Cleat	SA240GR304	
Manhole	NA	
SG/LG	NA	
Surface Finish	Internal Surface: SS :Pickling MS /CS : As per MDS	
	External Surface: SS :Pickling MS /CS : As per MDS	
Nozzle Schedule	As per MDS	
Qty	10 Nos	



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 4 OF 14

2) FLASH VESSEL

Tag No: FV-51203/51205/51308/51505/51602

Process Details:	Shell	Limpet
Fluid	Client to confirm	Client to confirm
Operating Temperature (°C)	180	280
Design Temperature (°C)	250	300
Operating Pressure (bar g)	1.5/FV	6
Design Pressure (bar g)	3.5 / Full Vacuum	20/FV
Hydro Test Pressure (bar g)	4.55	26
Density (kg/m³)	Client to confirm	Client to confirm
Corrosion Allowance	1	NIL
Radiography (Shell/Dish)	100% + 10 % + all T Joints	100% + 10 % + all T Joints
Joint Efficiency (Shell/Dish)	0.85/1	1
Vessel Details :		
Design Code	ASME SEC VIII, DIV-1	
Tank Shape	Cylindrical	
Orientation	Vertical	
Working volume (KL)	8 KL	
Gross Volume (KL)	12 KL	
Shell Inside Diameter	2050 mm	
Shell height TL to TL	3200 mm	
Shell thickness	10 mm (Nom)	
Top dish thickness	12 mm (Nom)/ 10.8 (Min)	
Bottom Dish thickness	12 mm (Nom)/ 10.8 (Min)	
Type of dish ends	Tori spherical	
Limpet		
Limpet OD mm	60	
Limpet pitch	90	
Limpet thickness mm	4 (Nom)	
No of turns	~11 on shell & ~10 on dish	

**TECHNO COMMERCIAL
OFFER FOR COLUMN &
CONSENSERS**

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 5 OF 14

Material of Construction	Shell / top dish / RF PAD	SA240GRSS316
	Body Flanges	NA
	Nozzle Pipe	SA312TP316(S)
	Nozzles Flanges up to 80NB	SA182F316
	Nozzles Flanges above 80NB	SA 516 GR 70 WITH SS316 OVERLAY
	Limpet Coil	SA240GR304
	Limpet Nozzle Pipe	SA312TP304 (S)
	Limpet Nozzle Flange	SA182F304 (SORF)
	Gasket	PTFE
	Int Bolt & Nuts	SS316
	Ext Fastners	SA193GRB7/SA194GR2HWITH HDGI
	Support	IS2062GRA WITH SS304 PAD
	Lifting Lug	IS2062GRA WITH SS304 PAD
	Internal	SA240GR316
	Insulation	IN CLIENTS SCOPE
Manhole	Earthing Boss	SS304 WITH BOLT
	Insulation Cleat	SA240GR304
SG/LG	NA	
Surface Finish	Internal Surface: SS :Pickling MS /CS : As per MDS	
	External Surface: SS :Pickling MS /CS : As per MDS	
Nozzle Schedule	As per MDS	
Qty	05 Nos	

3) MAIN CONDENSER

TAG NO: C-
51203/51205/51208/51302/51308/51407/51505/51602/51801/51805/51810/51801/51102/
51103/51906/51908

Process Details	Shell Side	Tube Side
Service	Client to confirm	Client to confirm
Flow rate (Kg/hr)	Client to confirm	Client to confirm
Operating Temperature in (°C)	150/165	40/35
Design Temperature in (°C)	200	100/-18
Operating Pressure in (kg/cm2g)	3.5/FV	3.5
Design Pressure in (kg/cm2g)	1 TORR	7
Test Pressure (kg/cm2g)	4.55	9.1
Viscosity (cP)	Client to confirm	Client to confirm
Density (Kg/m3)	Client to confirm	Client to confirm
No. of Passes per shell	1	4
Corrosion Allowance in mm	NIL	1.6

MECHANICAL DETAILS

Design Code	ASME SECTION VIII DIV.1& TEMA Class C
Joint Efficiency (Shell/Head)	0.85/1.00
Radiography (Shell/Head)	10% Spot + all T joints/100%

CONSTRUCTIONAL DETAILS

Heat Exchanger Area	20 SQM
Main Shell Shape/Orientation	Cylindrical/Horizontal
Flange Pressure Rating	ANSI B 16.5
Main Shell Inside Diameter (in mm)	550
Thickness (in mm)	5
Main Shell St Height	3000
Bonnet Shell Dia (in mm)	550
Thickness (in mm)	5
Bonnet Dish Thickness (in mm)	10 (Nom)/9 (Min)

MOC

Shell	SA 240 GR 316
CHANNEL SHELL	SA 516 GR 70
DISH ENDS	SA 516 GR 70



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 7 OF 14

GIRTH FLANGE	SA 516 GR 70
Nozzle NECK SHELL SIDE	SA312 TP 316 (S)
Nozzle NECK CHANNEL SIDE	SA106GRB
Nozzles Flanges up to 80NB SHELL SIDE	SA 182 F 316 (SORF)
Nozzles Flanges above 80NB SHELL SIDE	SA 516 GR 70 WITH SS316 OVERLAY
Nozzles Flanges up to 80NB CHANNEL SIDE	SA 516 GR 70 WITH SS316 OVERLAY
Nozzles Flanges above 80NB CHANNEL SIDE	SA 516 GR 70 WITH SS316 OVERLAY
RF PAD SHELL SIDE	SA 240 GR 316
RF PAD CHANNEL SIDE	SA 516 GR 70
Pass partition Plate	SA 516 GR 70
Coupling / Plugs	SA 182 F 316
Insulation cleats SHELL SIDE	SA240 GR 316
Insulation cleats Channel SIDE	SA 516 GR 70
Gasket	PTFE
Nut/Bolt	SA193 GR B7 / SA 194 GR 2H WITH HDGI
Insulation	In client's scope
Baffles	SA 240 GR 316
Tie Rods	SS316
Tubesheet	SA 240 GR 316
Tubes	SA 213 TP 316 (S)
Spacer Tubes	SA 312 TP316 (WLD)
Expansion Bellow	SS316
TUBE DETAILS	
No of Tubes (Nos)	84
Tube OD (in mm)	25.4
Tube Thickness (in mm)	2.11
Pitch	Triangular
Tube Length (in mm)	3000
Surface Finish Internal / External	SS: Pickling (As per MDS) MS: Cleaning & Painting (As per MDS)
Nozzle schedule	As per MDS
Quantity	15 Nos



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 8 OF 14

4) VENT CONDENSER

**TAG NO: VC-
51203/51205/51208/51302/51308/51407/51505/51602/51801/51805/51810/51102/51103
51906/51908**

Process Details	Shell Side	Tube Side
Service	Client to confirm	Client to confirm
Flow rate (Kg/hr)	Client to confirm	Client to confirm
Operating Temperature in (°C)	40/-10	150/165
Design Temperature in (°C)	100/-18	200
Operating Pressure in (kg/cm2g)	3.5	1 TORR
Design Pressure in (kg/cm2g)	7	3.5/FV
Test Pressure (kg/cm2g)	9.1	4.55
Viscosity (cP)	Client to confirm	Client to confirm
Density (Kg/m3)	Client to confirm	Client to confirm
No. of Passes per shell	1	1
Corrosion Allowance in mm	NA	
MECHANICAL DETAILS		
Design Code	ASME SECTION VIII DIV.1& TEMA Class C	
Joint Efficiency (Shell/Head)	0.85/1.00	
Radiography (Shell/Head)	10% Spot + all T joints/100%	
CONSTRUCTIONAL DETAILS		
Heat Exchanger Area	5 SQM	
Main Shell Shape/Orientation	Cylindrical/Vertical	
Flange Pressure Rating	ANSI B 16.5	
Main Shell Inside Diameter (in mm)	300NB SCH 30	
Main Shell St Height (in mm)	1500	
Bonnet Shell Dia (in mm)	300NB SCH 20	
Bonnet Dish Thickness (in mm)	6 (Nom)/5.4 (Min)	
MOC		
Shell	SA106GRB	
CHANNEL SHELL	SA 312 TP 316 (S)	
DISH ENDS	SA20GR316	
GIRTH FLANGE	SA516 GR 70 WITH SS316 OVERLAY	
Nozzle NECK SHELL SIDE	SA106GRB	



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 9 OF 14

Nozzle NECK CHANNEL SIDE	SA312 TP 316 (S)
Nozzles Flanges up to 80NB SHELL SIDE	SA 105
Nozzles Flanges above 80NB SHELL SIDE	SA 105
Nozzles Flanges up to 80NB CHANNEL SIDE	SA 312 TP 316 (S)
Nozzles Flanges above 80NB CHANNEL SIDE	SA 516 GR 70 WITH SS316 OVERLAY
RF PAD SHELL SIDE	SA 516 GR 70
RF PAD CHANNEL SIDE	SA 240 GR 316
Insulation cleats SHELL SIDE	SA 516 GR 70
Insulation cleats Channel SIDE	SA240 GR 316
Gasket	PTFE
Nut/Bolt	SA193 GR B7 / SA 194 GR 2H WITH HDGI
Insulation	In client's scope
Baffles	SA 240 GR 304
Tie Rods	SA 479 GR 304
Tubesheet	SA 240 GR 316
Tubes	SA 213 TP 316
Spacer Tubes	SA 312 TP304 (WLD)
Expansion Bellow	SS316
TUBE DETAILS	
No of Tubes (Nos)	43
Tube OD (in mm)	25.4
Tube Thickness (in mm)	2.11
Pitch	Triangular
Tube Length (in mm)	1500
Surface Finish Internal / External	SS: Pickling (As per MDS) MS: Cleaning & Painting (As per MDS)
Nozzle schedule	As per MDS
Quantity	15 Nos



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 10 OF 14

5) DISTILLATION COLUMN:

Tag No: DC-51203/DC-51205/DC-51208/DC-51302/DC-51308/DC-51407/DC-51505/DC-51602/DC-51801/DC-51505/DC-51801/DC-51102/DC-51103/DC-51906/DC-51908

Process Details:

Fluid	Client to confirm
Operating Temperature (°C)	170
Design Temperature (°C)	250
Operating Pressure (kg/cm ² g)	1.5/0.5TORR
Design Pressure (kg/cm ² g)	3.5/FV
Hydro Test Pressure (kg/cm ² g)	4.55
Density (kg/m ³)	CLIENT TO CONFIRM
Corrosion Allowance	SS-1.0/CS-NIL
Radiography (Shell/Dish)	100% + 10 % + all T Joints
Joint Efficiency (Shell/Dish)	0.85

Vessel Details :

Design Code	ASME SEC VIII, DIV-1	
Tank Shape	Cylindrical	
Orientation	Vertical	
Shell Inside Diameter	800 mm	
Shell height TL to TL	23800 mm	
Shell thickness	6 mm (Nom)	
Packing support ring	Support ring for packing (Only support considered, column internals & packing will be in client's scope)	
Material of Construction	Shell / top dish / RF PAD	SA240GRSS316
	Body Flanges	SA 516 GR 70 WITH SS315 OVERLAY
	Nozzle Pipe	SA312TP316(S)
	Nozzles Flanges up to 80NB	SA182F316
	Nozzles Flanges above 80NB	SA 516 GR 70 WITH SS316 OVERLAY
	Handhole Neck	SA 240 GR 316
	Hamdhole Flange	SA 516 GR 70 WITH SS316 OVERLAY



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 11 OF 14

	Handhole Cover	SA 516 GR 70 WITH SS316 Liner
	Gasket (SHELL)	PTFE
	Support internal	SA240 GR 316
	Int Bolt & Nuts	SS316
	Ext BOLTS & NUTS	SA193GRB7/SA194GR2HWITH HDGI
	Stiffners	SA 240 GR 316
	Support	IS2062GRA WITH SS316 PAD
	Lifting Lug	IS2062GRA WITH SS316 PAD
	Internal	SA240GR316
	Insulation	IN CLIENTS SCOPE
	Earthing Boss	SS304 WITH BOLT
	Insulation Cleat	SA240GR316
	SG/LG/FLANGE/COVER	SA240 GR 316
	Packing	IN CLIENTS SCOPE
Hand hole	250 NB	
SG/LG	100 NB_SS316L	
Surface Finish	Internal Surface: SS :Pickling MS /CS : As per MDS	
	External Surface: SS :Pickling MS /CS : As per MDS	
Nozzle Schedule	As per MDS	
Qty	15Nos	

Revision History

Revision	Comments / Changes	Commented By	Submission Date
R0	As per enquiry & MDS commercial submitted	Mr MG Attar/Ms. Subhasini S	15-01-2026



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 12 OF 14

GENERAL NOTES

-) The vessel shall be provided with all the nozzles mentioned in the MDS.
-) The nozzles orientation/size/projection should be confirmed by Client before the order finalization.
-) Compatibility of the MOC with in-tank contents of the reactor shall be in Client's Scope.
-) **Insulation if any will be in client's scope.**
-) **Internal packing / trays will be in client's scope.**
-) **Ladder & platform not considered in above offer.**
-) **Installation & commissioning in client's scope.**
-) **Above shared offer is as per DRAWINGS shared by client.**
-) **Process guarantee will be in client's scope. Mechanical guarantee will be in ACME scope.**
-) **Transportation cost in above offer is considered as per quantity shared. In case of any change in quantity will lead to change in transport costing as per required quantity.**
-) All material test certificates shall be provided. Material shall be of Non-Chinese origin.
-) Third party inspection shall be in Client's scope.
-) Site operation guarantee for complete package will be 12 months from the date of installation or 18 months from the data of dispatch whoever is earlier.
-) Any post order requirement or change in the design other than stated in the offer may lead to price implication.
-) All vessels shall be provided as DRAWING



ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)

TECHNO COMMERCIAL OFFER FOR COLUMN & CONSENSERS

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 13 OF 14

PRICE SUMMARY

SR. NO	ITEMS	Capacity	MOC	PRICE EACH	QTY	TOTAL PRICE
01	C-51203/C-51205/C-51208/C-51302/C-51308,C-51407/C-51505/C-51602/C-51805/C-51810/C-51801/C-51102/C-51103/C-51906/C-51908	20.0 M2	CS+SS316	11,78,000/-	15	1,76,70,000/-
02	DC-51203/DC-51205/DC-51208/DC-51302/DC-51308/DC-51407/DC-51505/DC-51602/DC-51801/DC-51505/DC-51602/DC-51801/DC-51102/DC-51103/DC-51906/DC-51908	800 ID	SS316	32,72,000/-	15	4,90,80,000/-
03	FV-51203/FV-51205/FV-51308/FV-51505/FV-51602	8.0M3	SS316	26,78,000/-	5	1,33,90,000/-
04	FV-51208/FV-51302/FV-51407/FV-51801/FV-51805/FV-51810/FV-51102/FV-51103/FV-51906/FV-51908	6.0M3	SS316	23,51,000/-	10	2,35,10,000/-
05	VC-51203/51205/51208/51302/51308/51407/51505/51602/51801/51805/51810/VC-51102/VC-51103/VC-51906/VC-51908	5.0 M2	CS+SS316	3,60,000/-	15	54,00,000/-
TOTAL FOR (INR)					60	₹ 10,90,50,000/-
IN WORDS: RUPEES TEN CRORE NINTY LAKHS FIFTY THOUSAND ONLY						

Commercial Terms

Price basis	: FOR, Mahad
Forwarding	: Inclusive
Insurance & Freight	: Inclusive
Taxes & Duties	: 18% GST Extra at actual.
Delivery Time	: 12~20 weeks from the date of drawing approval as per the priority of the equipment.
Payment Terms	: 50% advance, 50% against PI before dispatch
Validity	: 10 Days.

**ACME PROCESS SYSTEMS PVT LTD, PUNE (INDIA)****TECHNO COMMERCIAL
OFFER FOR COLUMN &
CONSENSERS**

ISSUE:	VERSION : 0
DOCUMENT No.	APSPL/PSCL/1592/25-26/R0
DATE :	15-01-2026
PAGE No.	PAGE 14 OF 14

We trust this is in line with your requirements. Should you require any further information then please feel free to contact us.

Thanks & regards,

AABHA KULKARNI

SR. ENGINEER – Application & Sales

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ACME PROCESS SYSTEM PVT. LTD.